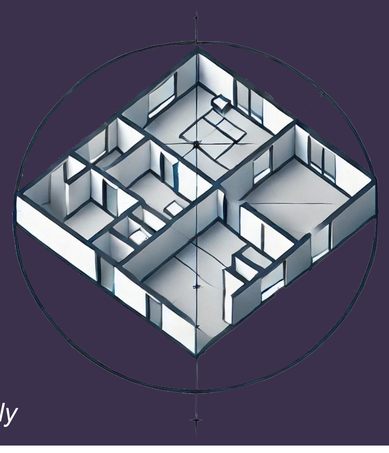
"Flow along the corridor"

A GAN-based floor plan generator



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01 Introduction

Generative design has recently become a highly prominent topic. In the past few years, machine learning tools have proven invaluable in creating more precise and diverse 2D or 3D building models, enhancing user engagement in the design process. A review of past approaches highlights the use of techniques such as diffusion models, generative adversarial networks (GANs), and graph neural networks (GNNs), all of which aim to enable computers to comprehend and reconstruct the complexities of building design.

This research builds on this foundation by developing a method for machine learning to understand, learn, and reconstruct building plan information, ultimately creating a generative tool. The model allows users to customize the generation process, focusing on essential circulation lines (i.e., corridor spaces) to produce functional building plans. After validating the feasibility of this methodology, the study offers a viable solution for the subsequent generation of architectural 3D spaces.

Keywords: Generative Design, 2D GAN, Spatial Relationship, Graph Theory, Methodological Exploration

03 Resaech Objectives

Comparison with previous studies, this research will solve:

1, This research attempts to find **a more precise method** to capture building features and learn from it (both applies to 2d and 3d).

2, Not only single apartment, but also explore the possibility to **generate public buildings.**

3, In addition to meeting the basic conditions for rational building generation, this study will also introduce **'circulation'**, which is an important element of public buildings.

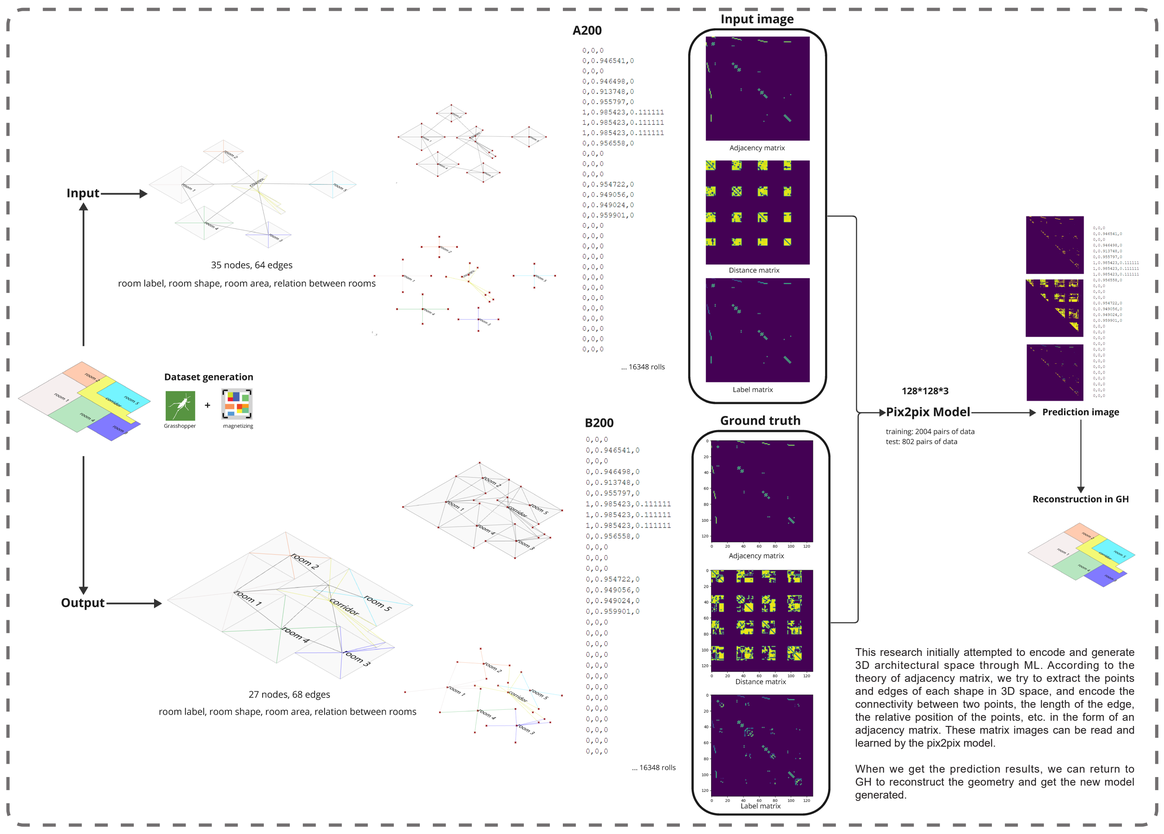
02 Literature Review

Overview for 2D/3D Generation Methodology

Title	Figure	Method	User input	Output	Output format	Scope of application
ArchiGAN[06]		Pix2Pix	boundary	room configuration with detailed furniture	Raster	both single and large apartment
Architectural layout design through deep learning and agent-based modeling: A hybrid approach[10]	Heat Map Layout Design	Pix2Pix + cGAN	boundary/room relationship/room area	room configuration	Raster	single apartment
House-GAN++[11]		GANs + Convolutional message passing	room relationship	room configuration with door position	Vector	single apartment
Graph2plan[07]		GNN + GAN	boundary/room relationship/room location	room configuration with door and window position	Vector	single apartment
floorplanGAN[12]	Bubble Diagram Preliminary Layout	GAN	room function/area	room configuration	Vector	single apartment
HouseDiffusion[13]		Diffusion Model	room relationship	room configuration with door position	Vector	single apartment
Graph-Constrained House Generation[09]	Input Graph Learned Attention GTGAN Ground Truth	graphTransformer	room relationship	room configuration	Vector	single apartment
Our work		Pix2pix	corridor line/room area/room relationship(optional)	room configuration	Vector	public buildings

Title	Figure	Method	Training input	Output	Output format	Scope of application
A generative architectural and urban design method through artificial neural networks[01]		ANN	Encoded points and surfaces	points and UV surface model	Vector	high-rise building volume
Synthesis and generation for 3D architecture volume with generative modeling[02]		3D GAN	voxel matrix	3D voxel model	Voxel	high-rise building volume
Building massing generation using GAN trained on Dutch 3D city models[05]		2D GAN	block surrounding	new buildings in the void space	Voxel	building volume
Urban Layout Workflow Utilizing Generative Adversarial Network (GAN)[03]	Result Bitmap	2D GAN	layers of image with different information	3D urban block model	Vector	block volume
Building-GAN[04]	Input Program Ratio Program Graph Irregular Voxel Grid Voxel Graph	GNN+pointer-based cross-modal modules	graph	Voxel building model with internal spatial divisions	Voxel	single public building with internal spatial divisions

04 Methodology Workflow

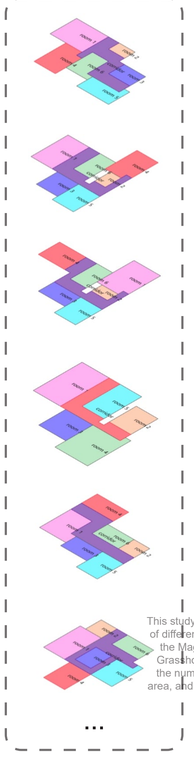


This research initially attempted to encode and generate 3D architectural space through ML. According to the theory of adjacency matrix, we try to extract the points and edges of each shape in 3D space, and encode the connectivity between two points, the length of the edge, the relative position of the points, etc. in the form of an adjacency matrix. These matrix images can be read and learned by the pix2pix model.

When we get the prediction results, we can return to GH to reconstruct the geometry and get the new model generated.

05 Dataset generation

Grasshopper Generate Dataset

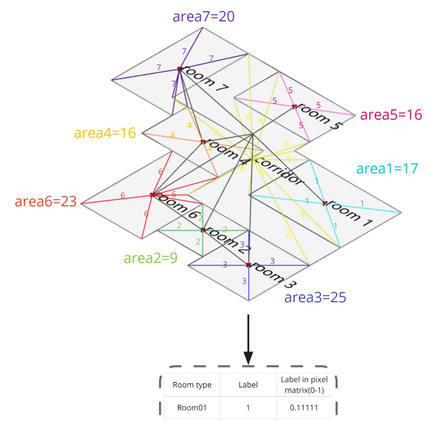


This study produced thousands of different floor plans through the Magnetizing plug-in in Grasshopper. They differ in the number of rooms, room area, and relationship between rooms.

Embed Features

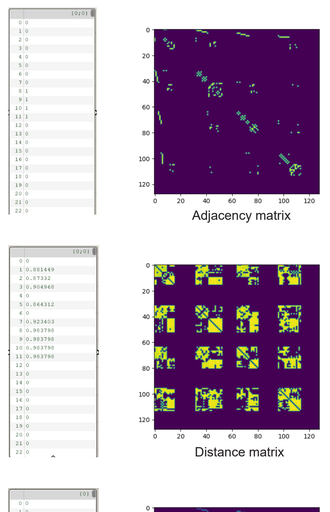
Here we use nodes and edges to depict a building information.

It includes: room label, room shape, room area, relation between rooms.



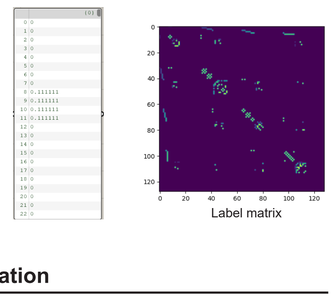
Room type	Label	Label in pixel matrix(0-1)
Room01	1	0.11111
Room02	2	0.22222
Room03	3	0.33333
Room04	4	0.44444
Room05	5	0.555556
Room06	6	0.666667
Room07	7	0.777778
Corridor	8	0.888889
Others	9	1

Room relation/room geometry
Nodes connections(yes/no)

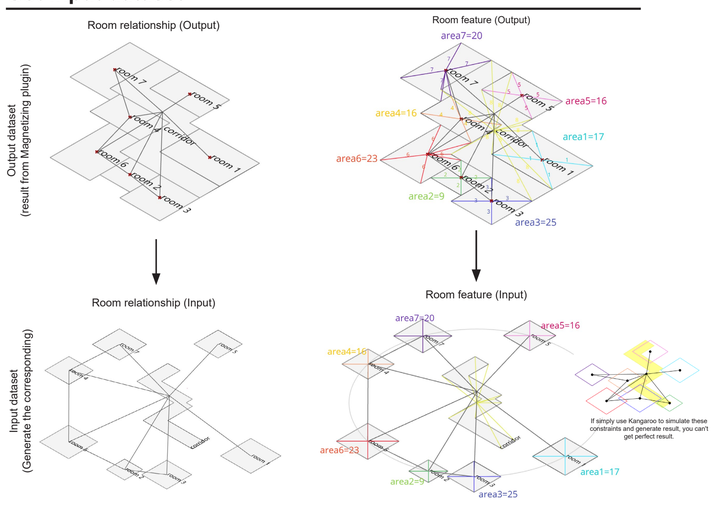


Room relation/room geometry
Nodes distance/ edge length

Room label
edge label



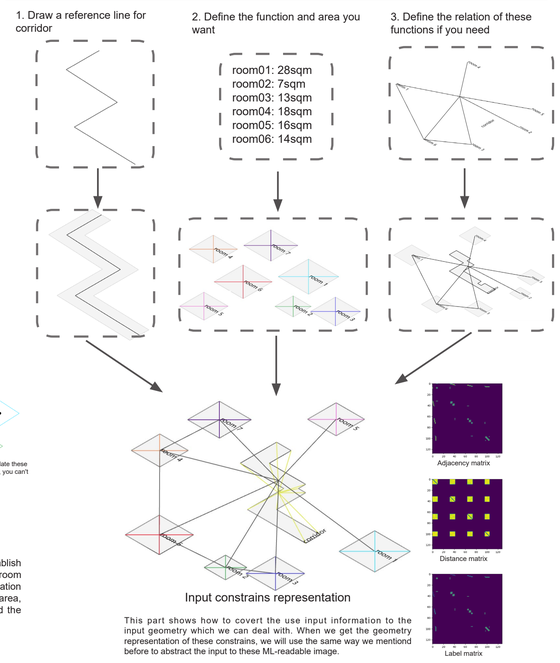
Got input dataset



Translation of relationships between rooms: Connect scattered rooms in the input data set using the same connection method as the output data set

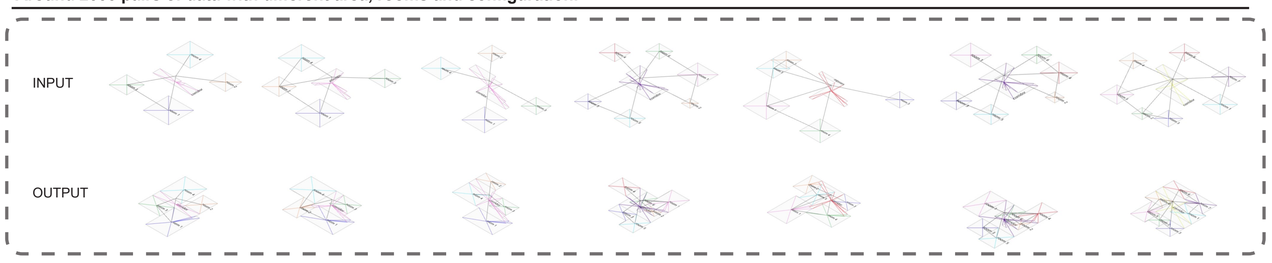
The input data set should have the same information as the output data set and establish a relationship. We can obtain information about corridor shapes, room areas, and room relationships from the thousands of generated floor plans. We re-express this information centered on the corridor: the input room is transformed into a square with the same area, the rooms are connected with the same line to express the room relationship, and the shape of the corridor is retained... From this, we get the shape of the input data set.

User Input representation

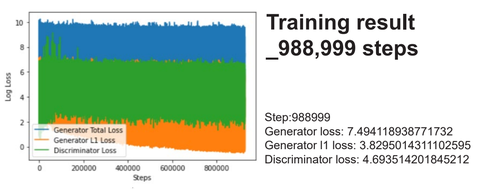


This part shows how to covert the use input information to the input geometry which we can deal with. When we get the geometry representation of these constrains, we will use the same way we mentiond before to abstract the input to these ML-readable image.

Around 2000 pairs of data with different area, rooms and configuration.



06 Model Training



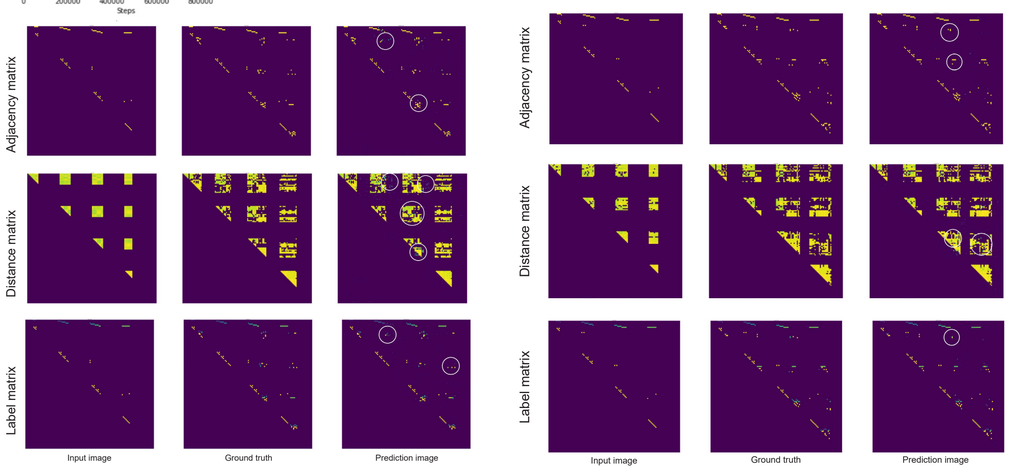
Training result _988,999 steps

Step:988999
Generator loss: 7.494118938771732
Generator l1 loss: 3.8295014311102595
Discriminator loss: 4.693514201845212

Prediction Result

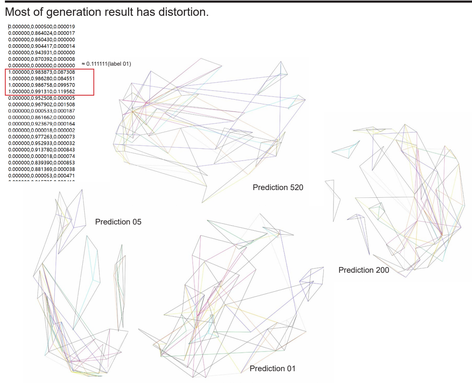
We analyzed the results of the two training sessions (short and long) separately. In the long time training, we trained approximately 988,999 steps. Comparing the data at the beginning of training, we can see that Generation loss and Generation l1 loss continue to decrease, and Discriminator loss continues to increase, which shows that we are moving in the right direction.

When we plot the prediction results, we can see that the adjacency matrix and label matrix still have relatively small errors. The error in the distance matrix prediction results is also significantly reduced. Most of the pixels have the correct color, with only two obvious errors.



Reconstruction Result_433,999 steps

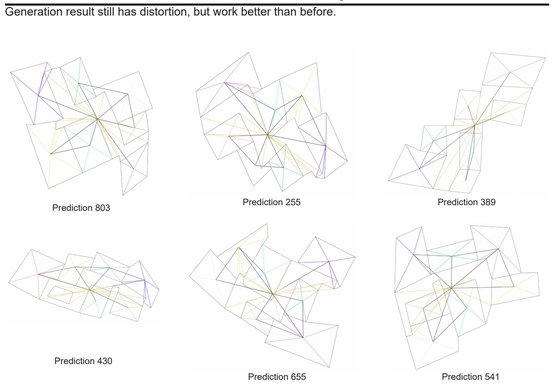
Most of generation result has distortion.



This part shows the reconstruction result in grasshopper. After training for about 400,000 steps, we did not get a good result.
Almost most of the reconstruction results have large distortions. They do not form a continuous flat floor plan. This result is obviously unusable. But by observing the results, we can know that the contents of the adjacency matrix and the label matrix are better predicted to a certain extent, They conform to the rule of appearing in pairs during input. And from GH we can recognize that different colors are scattered together.

Reconstruction Result_988,999 steps

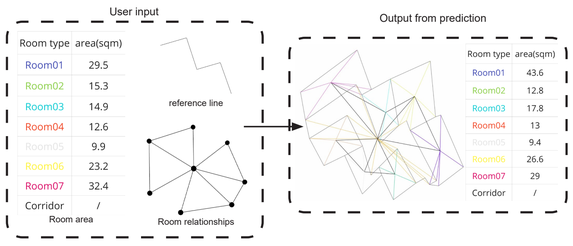
Generation result still has distortion, but work better than before.



When we reconstruct the geometry in Grasshopper, we can clearly see that the prediction results this time have significantly improved. The result though is still not a flat floor plan and there is still a distortion in the shape of the room within it. But the logic of their layout is already discernible in these floor plans. Colored edges are distributed throughout the interior of each room, and there is no color confusion. The connection between the rooms has also been restored very well.

07 Conclusions

Conclusion



The method in this study enables ML to accurately capture the geometric relationships in building plans, including room foot points, center points, and the edges that define room shapes. Most linear geometric shapes can theoretically be represented by nodes and edges along their boundaries. Thus, compared to previous methods, our deeply trained model should more accurately reconstruct complex spaces.

The Pix2Pix model's multiple channels allow for the learning of various layers of information. In this study, we introduced a new channel that identifies room labels using adjacency matrix theory, demonstrating its feasibility. This also suggests the potential to incorporate additional information layers to generate more complex models.

Limitation

1, Pix2pix HD Model can deal with the image size of 1024 dpi. This generator cannot generate too big or too complex 3D buildings which has more than 1024 feature nodes.

2, This method has relatively high requirements for the depth of model training. In the adjacency matrix, a small pixel error will cause a relatively large deviation. And obtaining an accurate distance matrix often takes longer to learn.

Future

1, The previous research has shown that this method is feasible. Next, we can increase the training time or find a dataset that is more consistent with the actual situation and retrain the model to make the results more ideal.

2, Not only for the floor plan. How to use this method in the generation of 3D space requires further exploration.

3, It allows us to add more channels for different information. So, change or add more different constraints(sun hours, surrounding), let users control or interact with generator more intelligent.

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